Network Modeling and Analysis in Python

Today's goal • 3/3 items

Module 1
Connectivity in Networks

Welcome to the Course

- Welcome to Network Modeling and Analysis in Python
  Video • 5 min
- MADSwPy Certificate Roadmap
  Reading • 10 min
- Course Syllabus
  Reading • 10 min
- **Introduction to Jupyter Notebook**
  Reading • 10 min
- Meet Other Learners
  Discussion Prompt • 10 min
- Help Us Learn About You
  Reading • 10 min

Homophily and Assortativity

- Homophily and Assortativity
  Video • 16 min

Structural Holes

- Structural Holes
  Video • 12 min

K-core Decomposition

- K-core Decomposition
  Video • 8 min
- Knowledge Check: Connectivity in Networks
  Practice Assignment • 35 min

Module 1 Wrap-Up

- Demo: Connectivity NetworkX Tutorial
  Video • 6 min
- Module 1 NetworkX Tutorial
  Lab • 1h
- Module 1 Assignment
  Graded Assignment • 30 min
- Module 1 Programming Assignment: Connectivity in Networks
  Programming Assignment • 3h
- Optional Readings and Resources
  Reading • 10 min

Optional Lesson: Review of Key Concepts

- Review of Key Concept
  Reading • 10 min
- Network Definitions and Vocabulary
  Reading • 10 min
- Network Connectivity
  Reading • 10 min
- Influence Measure and Network Centralization
  Reading • 10 min
- Network Evolution
  Reading • 10 min
- Optional-Review of Key Concepts
  Practice Assignment • 30 min

Module 2
Community Structure in Networks

Module 3
Network Generative Models

Module 4
Models of Diffusion in Networks

Introduction to Jupyter Notebook

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This introduction provides you with the basic instructions on how to effectively navigate and use Jupyter Labs on Coursera, ensuring you can make the most of your learning experience in this powerful space.

What is Jupyter Labs?

In this course, you will execute code directly on the Coursera platform using Coursera Labs. The labs are set up in an environment that leverages Jupyter Notebooks to present **assignment instructions** and **read-only pre-written code for execution**. This environment includes all necessary packages to run the pre-written code and most types of code relevant to the course. It is a ready-to-use workspace for learners to engage with the course material.

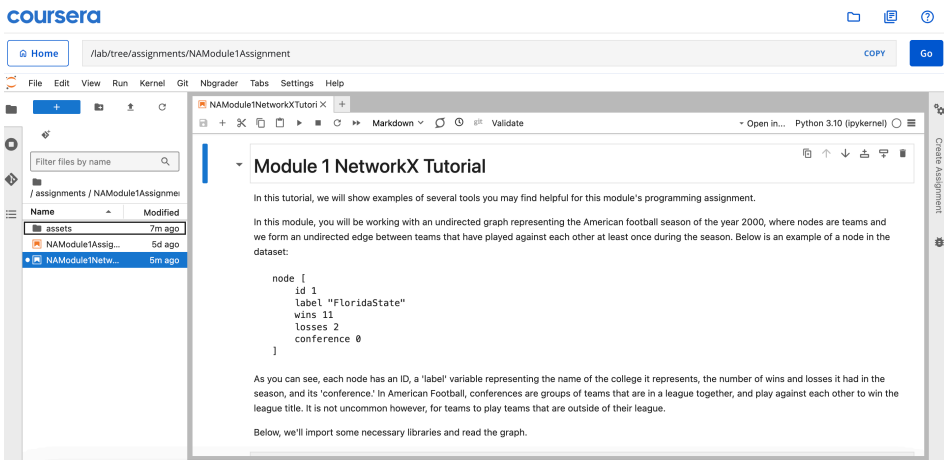
Note that as this platform is hosted in a shared cloud environment, some code may take longer to execute than it would if learners would execute the code themselves on their local machine. The Coursera Labs environment provides a convenient workspace for learners to write and execute code in, however the resources available to execute learner code on are fairly limited. Please be aware that some code segments, particularly those that utilize machine learning libraries, may take 30 minutes or longer to execute in full.

Navigating Jupyter Labs

Jupyter lab space includes two key components as shown in Figure 1:

- **A file browser** located on the left-hand side, and
- **A notebook interface** located on the right

Figure 1. A screenshot of the Jupyter lab environment



In this course, you will work in the notebook for the most of the time, especially in the tabs ending with ".ipynb". This file extension stands for "Interactive Python Notebook". Within this interface, you will notice code cells and, sometimes, markdown (text) cells. Here's how to work with them:

1. **Code Cells** contain Python code that you can execute to see the results immediately.
2. **Markdown (Text) Cells** contain text, instructions, or explanations. They are not meant for executing code but they provide important context and information.

To effectively use the programming exercises included in this course, first review the "implementation" document that we have provided before each programming exercise. This document explains different parts of the code and the output.

Run Pre-Written Code Cells

After reviewing the Implementation document, run the code in the Jupyter notebook. To run the code cells:

1. Select the Cell: Click on the cell to select it.
2. Execute the Cell: Use any one of the following methods to execute the selected cell.
3. Use one of these methods:
   - Press "Shift + Enter" keys to run the cell and move to the next one.
   - Press "Ctrl + Enter" to run the cell and stay in the same cell.
   - Click the "Run" button in the toolbar (represented by a "play" icon).

Running Code Cells

Some cells allow you to modify the code and experiment with your own implementations. Follow the assignment instructions, you will be asked to paste your code into these cells and run them to see the results.

1. Edit the Code: Click on the editable cell and modify the code as per the instructions.
2. Run the Cell: Use any one of the following methods to execute the selected cell.
   - Press "Shift + Enter" to execute the modified code and move to the next cell.
   - Press "Ctrl + Enter" to execute the code and stay in the same cell.
   - Click the "Run" button in the toolbar (represented by a "play" icon).

Mark as completed